NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Interação Pessoa-Máquina

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Music Maker Hub

Stage 2: User and task analysis

Project's logo

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1. Problem Description

Music Maker Mod is a Minecraft mod that allows players to create and play music inside Minecraft. Currently, players lack a convenient and organized way to share, discover, and evaluate *.sheet* files (files that can be imported/exported through the mod itself). Files are often exchanged through Discord, where messages can become difficult to find and organize, while previewing a sheet requires downloading it and opening it in Minecraft first, making it harder for users to decide whether they want to use it.

2. User Analysis

2.1 Target Users

The project targets Music Maker Mod players, creators, and community members who want to publish, discover, preview, download, and discuss music sheets.

2.2 User Classes

We have identified two user classes for this project:

1. Regular users: Can browse, preview, upload, download, comment, like, save, rate and submit feedback
2. Administrators: Can perform regular user actions, plus moderate content, review reports, manage uploads, and handle administrative tasks.

3. Task Analysis

3.1 Task: Search Music

The main objective of the “Search Music” task is to help users find published music sheets that match their interests. Users can search by text, narrow the results by instruments, compare sheets using different sorting filters and open the sheet’s page for more details.

The task has the following precondition:

- Published sheets must be visible in the website’s catalog. Users don't need to be logged in to search or browse them.

The main sub-tasks are:

1. Enter a search term in the search bar and submit it. The search checks sheet titles, authors, description, tags and part names.
2. Browse published sheets without entering a search term.
3. Select one or more instruments to narrow the results. When selecting many instruments, a sheet must contain all of them to match.

4. Sort the display results by newest, popularity, likes, saves, views, rating, title, note count or length.
5. Compare results using their titles, authors, instruments and duration.

A search may return no matches if the terms are too specific, they don't exist or the selected instruments do not occur together in a sheet. Incomplete sheet information can also prevent an otherwise relevant result from matching.

3.2 Task: Upload and Download Music

The main objective of the "Upload and Download Music" task is to allow users to upload their own creative works and download anything they enjoy inside the website. In the upload setting, users can first preview their content before uploading as well as change any information about the song they want to (instruments, title, description, parts, etc). Regarding downloads, the user can download any existing music sheet he desires, as well as choose between the full set (equivalent to all the parts that compose a sheet), or just specific parts (e.g: just the saxophone part from a trio ensemble).

A couple preconditions must be met to enable this functionality, and they differ for uploading and downloading:

- For downloading, the sheet must already exist on the website and be accessible to the visitor.
- For uploading, the user must be signed in and select at least one valid *.sheet* file.
- For uploading, the user must use sheets that have the same tempo, otherwise they will be rejected (for this to make sense, let's make an analogy: A car cannot properly drive if it has 3 normal tires and a truck tire. Music should also follow that rule of consistency). A new part added to an existing set must also match that set's tempo (BPS).
- For uploading, a title and author must be provided.

The task consists of the following sub-tasks. Anytime a preview is mentioned, we are talking in regard to the upload feature:

1. Select one or more *.sheet* files and generate a preview locally, before sending them to the server.
2. Play, pause and inspect the combined preview.
3. Add parts to the preview.
4. Remove parts from the preview.
5. Edit the information of the sheet (title, author, description) while previewing.
6. Change a part's display name or select an instrument override for its website preview.
7. Download the full set of parts that compose a sheet in a *.zip* folder containing all the *.sheet* files.
8. Download specific parts of a sheet, either individually or together.

Possible exceptions could include an invalid file, mismatched BPS values, missing required exceptions or network issues. A request download may fail if the sheet has been removed or is no longer accessible.

3.3 Task: Play and Edit Music

The main objective of the “Play and Edit Music” task is to let users listen to music sheets through an interactive piano-roll view, and to allow users to download the sheets to use them with the Minecraft mod. Users can choose which instrument parts to hear and adjust each part’s volume. In this task, editing refers to an uploader or admin changing sheet details and managing its parts.

Several pre-conditions must be met to enable this functionality:

- Playback requires an accessible music sheet with valid notes, timing and instrument data, together with a browser that supports the required audio features.
- Changing a sheet’s details or parts requires the user to be its uploader or an admin.

The main subtasks are:

1. Allow users to open the sheet page and view its notes in a piano roll, with note positions showing pitch and timing, as well as a scrollbar showing progress.
2. Allow users to play, pause, seek through and return to the beginning of the piece.
3. Allow users to select one or multiple instrument parts for playback and adjust each part’s volume.
4. Allow users to download an individual part as a mod-compatible *.sheet* file or download multiple parts as a *.zip* folder
5. Where authorized, edit the sheet’s descriptive information and manage its uploaded parts.

Playback may fail if the browser does not support the required audio features or if the sheet data cannot be loaded. Instrument samples or downloads may be affected due to network issues. An editing attempt made by a user who lacks permissions must be rejected.

3.4 Task: Interact with Users and Music Contributions

The main objective of the “Interact with Users and Music Contributions” task is to allow users to socially interact with each other's music contributions. The intended features between the user class should support discovery of music, feedback and opinion exchange as well as the sharing of music. Interaction between the user and admin class is also part of the objective to report inappropriate content and to answer frequently asked questions.
Several pre-conditions must be met for enabling interaction, especially as it builds on the previously described tasks.

- User accounts must exist to interact with community elements.
- Users must be authenticated to interact and engage.
- The selected contribution being interacted with, rather it be a music or comment, must exist.

- The user has opened or selected the contribution they want to interact with.
- For reports, a moderation process must exist to receive and review the submitted report.

Coming to the sub-tasks, it is unlikely that a stringent process will emerge within this task. The task is rather composed of individual features and functionalities:

1. Allow a logged-in user to like or unlike a music contribution.
2. Allow a logged-in user to save or remove a contribution from their personal list.
3. Allow a logged-in user to submit or update a rating.
4. Allow a logged-in user to write a comment and display their username and timestamp.
5. Allow a logged-in user to respond to existing comments and continue discussions.
6. Allow a logged-in user to show appreciation for a useful or interesting comment.
7. Allow users to share a contribution through a link.
8. Allow a logged-in user to submit a report and explanation to the administrators.
9. Allow administrators to view submitted reports, open the related contribution and mark reports as reviewed.
10. Show users their saved, liked and commented in their account area.
11. Display public interactions counts, such as the number of views, likes, rating and saves and comments.

Exceptions include an expired sign-in session, or a sheet or comment that is no longer accessible. Reports may remain unresolved until an admin reviews it. Use is expected to be frequent, but this is an exception rather than a measured figure. Likes, ratings and saves are supposed to be frequent, while comments and reports take longer to complete.

4. Scenario Design

Scenario 1: Finding a part for rehearsal

Sofia needs a saxophone part for a Minecraft server performance that evening. She searches the library for sheets containing both piano and saxophone, listens to a result, and opens it. She selects the saxophone part, then downloads that part's *.sheet* file for the mod.

Scenario 2: Publishing an arrangement

Rui has made a piano-and-flute arrangement. He signs in, selects both *.sheet* files and listens to the local preview before uploading. He corrects the flute part's name, adds a title and description, and confirms the upload. Because both files have the same BPS value, the set can proceed to server scanning and publication; a mismatch would require Rui to replace one of the files.

Scenario 3: Taking part in the community

Inês opens a sheet that a friend shared with her. She signs in, saves it and leaves a comment showing her appreciation for the score. While reading its details, she notices that the author's credit appears incorrect. She reports the sheet and types an explanation. An administrator sees the reason in the Reports section, opens the sheet to investigate and marks the report as reviewed.

5. Interviews

After speaking with a couple potential users for the website, we gathered that the main functionalities needed are the ones that have already been talked about and that the group intends to move forward with.
The consensus between all users spoken with was that the website needed:

1) Allow users to play music sheets without having the need to download them first;
2) Allow users to upload their own works
3) Allow users to search through the existing sheets

There were some other features that were mentioned by some of the people interviewed in regard to having community features, mostly about allowing users to comment under sheets to either praise or make suggestions.